What we're building, and why

A plain-language guide to the ECE 473 project · for Bryan · November 3, 2026

We are building a **market maker** for Kalshi contracts on NBA games. A market maker is a program that always offers to buy and sell a contract, earning the small gap between its buy and sell prices. Ours is different from a generic one in a single way: instead of pricing from what other traders are doing, it prices from **its own forecast of who will win**, updated after every play. The whole project is a test of whether that forecast makes the market maker better.

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1. The contract and the market

Kalshi lists one contract per team per game, for example `KXNBAGAME-25JUN19OKCIND-IND` = "Indiana wins Game 6". A YES contract pays **$1 if the team wins, $0 if it loses**. It trades at a price between 0 and 1, and that price is the crowd's probability that the team wins. If Indiana YES trades at $0.34, the market thinks Indiana has a 34% chance.

Like a stock, the contract has an order book: a **bid** (highest price someone will pay) and an **ask** (lowest price someone will sell at). The midpoint is the **mid**. During the game, the price jumps every time something happens: a 3-pointer, a run, an injury.

> **THE TWO KINDS OF TRADER**
>
> A **taker** hits an existing bid or ask because they want to trade now. A **maker** posts bids and asks and waits. Makers earn the spread each time they get hit on both sides, but lose when a taker knows something they don't (for example, the taker saw the 3-pointer go in before the maker updated its price). Bürgi et al. (2025) found that on Kalshi makers earn better returns than takers, and that the crowd overprices longshots. That's the opening we're going after.

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2. The question we are answering

Every automated market maker needs a **center price** to put its bid and ask around. The usual choice is the market mid. Our claim: for NBA games there is a better center, a **calibrated live win probability** built from the score, the clock and how good the teams are. The project compares the two, with the same spread rules, fees and fill simulation, on games neither has seen.

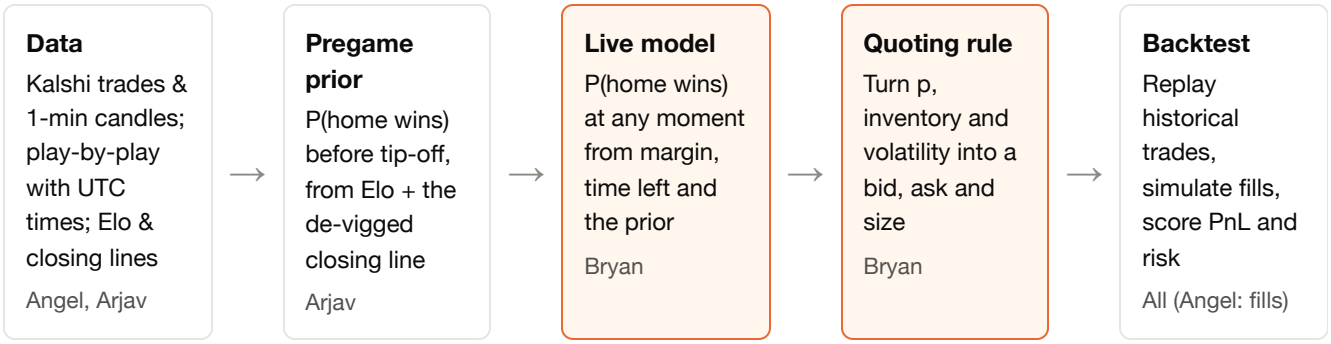
> **FORECAST QUESTION**
>
> Is our P(home wins) better than the Kalshi price? Scored with log loss, Brier score and calibration.

> **TRADING QUESTION**
>
> Does a market maker centered on our forecast make more money, with less risk, than one centered on the mid? Scored with PnL in a backtest.

The proposal also adds two weaker baselines so we can see *where* the value comes from: a **pregame-only** forecast that never updates during the game, and the vig-free **Pinnacle** sportsbook line.

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3. How the pieces fit together



Your two boxes are the orange ones. Everything upstream feeds you a number (p_{pre}) and a table (the play-by-play); everything downstream takes two functions from you (`p(state)` and `quote(p, inventory, state)`) and plugs them into the same engine every baseline runs through.

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4. The live model: Stern's Brownian motion

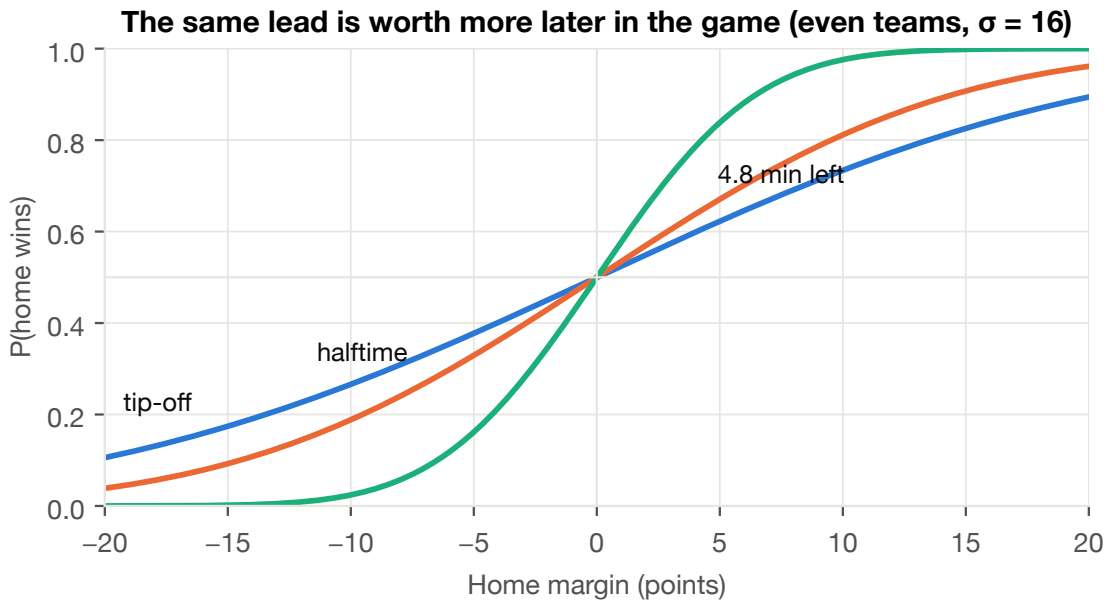
This is your first model and the baseline every later model has to beat. Stern (1994) made one simple assumption: the **score margin wanders like a random walk**. It drifts toward the better team, and it wiggles randomly as the two teams trade baskets.

Say the home team leads by m points with a fraction τ of the game left ($\tau = 1$ at tip-off, 0 at the buzzer). The rest of the game adds a random amount to the margin: on average $\mu\tau$ (the drift), with standard deviation $\sigma\sqrt{\tau}$ (the noise, which grows with the time remaining). The home team wins if the final margin is above zero, which gives a one-line formula:

$$\text{P(home wins)} = \Phi\left(\frac{m + \mu\tau}{\sigma\sqrt{\tau}}\right)$$

Symbol	Meaning	Where it comes from
m	Home margin right now	scoreHome – scoreAway from play-by-play
τ	Fraction of game left	seconds left ÷ 2880
σ	How noisy a full game's margin is, in points	**The only thing you fit.** Chosen to minimize log loss on 2015–24 games
p_{pre}	Pregame P(home wins)	Arjav's prior (Elo + closing line). Prototype: Kalshi mid at tip-off
μ	Expected full-game margin	$\mu = \sigma \cdot \Phi^{-1}(p_{pre})$, chosen so the formula returns p_{pre} at tip-off
Φ	Standard normal CDF	`scipy.stats.norm.cdf`

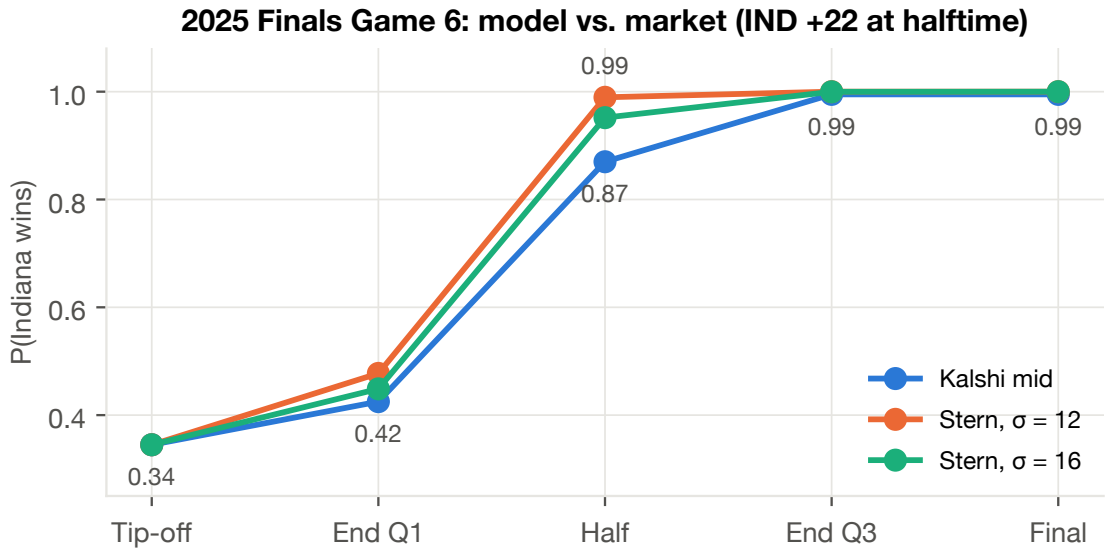
Two sanity checks fall straight out of the formula. At tip-off ($m = 0$, $\tau = 1$) it gives exactly p_{pre}. As time runs out ($\tau \to 0$) the denominator shrinks, so any lead pushes the probability to 1 and any deficit to 0. That is the whole intuition: **the same lead is worth more the less time is left.**



P(home wins) against the current margin at three moments of a game between even teams. A 10-point lead is 76% at tip-off but 99% with under five minutes left.

What happened on the one game we tried

We ran the model on 2025 Finals Game 6 (Indiana at home, Kalshi mid 0.345 at tip-off) with a guessed $\sigma = 12$. Direction right, confidence wrong: at halftime Indiana led by 22 and the model said 99%, while the market said 87%.



Quarter-end snapshots. Stern's own NBA estimate was closer to $\sigma \approx 16$; even that leaves a gap at halftime (0.95 vs. 0.87). One game proves nothing, so the next step is to check this across thousands.

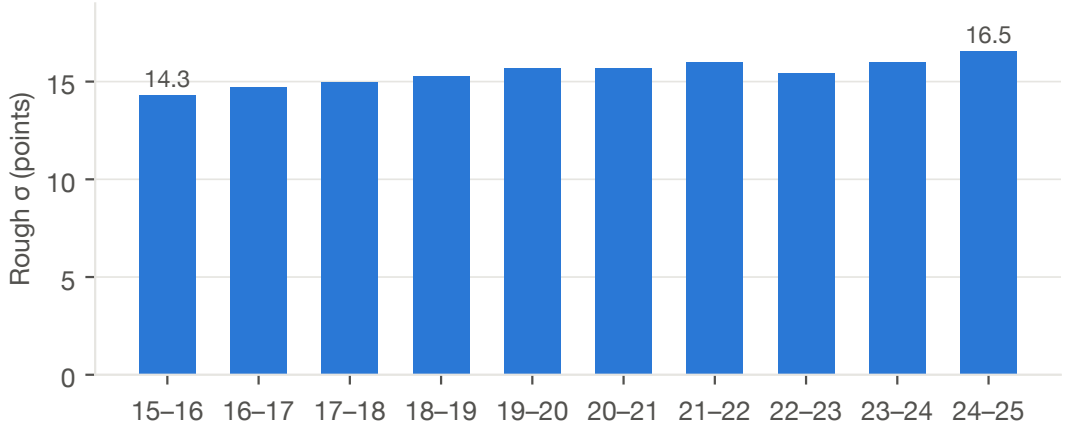
TWO POSSIBLE EXPLANATIONS, AND THE PROJECT TURNS ON WHICH IS TRUE

(a) σ = 12 is just wrong, so fitting it on real games will close the gap. **(b)** The market overprices comebacks (the favorite-longshot bias Bürgi et al. found). If (b) survives a proper fit, the market maker has an edge to harvest. Fitting σ is how you find out, which is why it is the first task on the Milestone II list.

What the data says about σ so far

From the 12,300 regular-season games we downloaded, the spread of the second-half margin change gives a rough σ per season. Two things stand out: it is **well above 12** everywhere, and it has **risen about 15% since 2015** as three-point shooting grew. So one σ pooled across all ten seasons would be too small for 2026 games; the 2024–25 validation season should decide how much the old seasons count.

Games got noisier: rough σ from the second-half margin change, by season



Rough σ = SD(final margin − halftime margin) ÷ √0.5. It ignores team strength, so it's a ballpark, not a fit. Home win rate fell from about 59% to 54% over the same period, which matters for the Elo prior.

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5. How we fit and score a forecast

The training table has one row every 15 seconds of game clock: margin, seconds left, p_{pre}, and whether the home team actually won (y = 1 or 0). The model gives a probability p for each row. We pick σ to minimize **log loss**, which punishes confident mistakes hard:

$$\text{log loss} = -\frac{1}{N}\Sigma\left[\, y \log p + (1-y)\log(1-p) \,\right]$$

Saying 0.99 and being wrong costs log(0.01) ≈ 4.6; saying 0.87 and being wrong costs 2.0. That is exactly the Game 6 question in numbers.

Metric	What it measures	Good value
Log loss	Overall forecast quality; punishes overconfidence	Lower. Beat the Kalshi mid's log loss on the same rows
Brier score	Mean of $(p - y)^2$; same idea, gentler on tails	Lower. 0.25 = coin flip
Calibration	Of the times we said 70%, did the home team win 70%?	Points on the diagonal of a reliability plot

The bar to clear is the logistic regression from Yeh et al. (2022), who showed ESPN's live forecast is no better than this model on four features: $m/\sqrt{\tau}$, $\text{logit}(p_{pre})\cdot\tau$, τ, and who has the ball. That is model 2 on your list; model 3 is a gradient-boosted tree, kept only if it beats model 2 on the validation season.

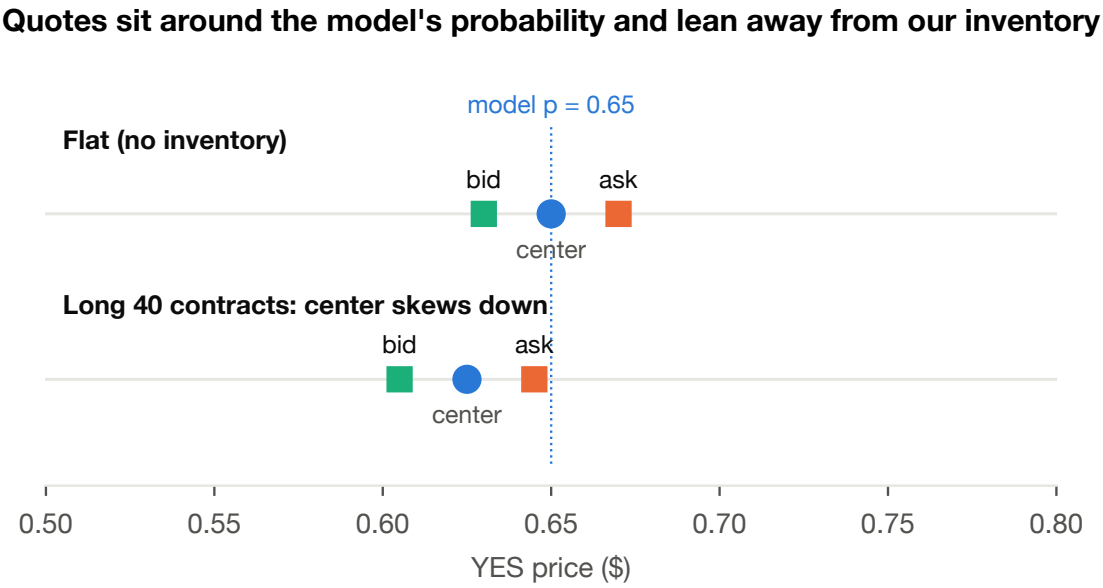
> **WHY THE SEASONS ARE SPLIT THE WAY THEY ARE**
>
> **2015–16 to 2023–24**: fit the live model. **2024–25**: choose between models and calibrate (nothing is fitted here). **2025 playoffs to Jan 2026**: tune the market maker's knobs on real Kalshi data. **Feb to Jun 2026**: the test, run once. Nothing from later ever touches something earlier. That's what makes a result believable.

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6. The quoting rule: turning p into a bid and ask

Once the model says p = 0.65, the market maker posts a bid a little below and an ask a little above, and waits. Four pieces are added one at a time, each checked on the tuning period before the next goes in.

1. **Fixed half-spread** δ around the center, fixed size, and an inventory cap. Quote bid = $c - \delta$, ask = $c + \delta$.
2. **Inventory skew** (Avellaneda–Stoikov, 2008). If we are long q contracts, we want to sell more than buy, so we shift the center down: $r = c - \gamma \cdot q \cdot p(1-p)$. The $p(1-p)$ term is the variance of a 0/1 payout, so the skew is strongest around 50/50 and vanishes near 0 or 1.
3. **Volatility-based spreads.** Widen when p is about to move a lot: late in close games, and for a few seconds after a scoring play. The Stern model hands us the amount for free (next page).
4. **Blended center** in log-odds: $w \cdot \text{logit}(p_{model}) + (1 - w) \cdot \text{logit}(mid)$. w is tuned; w = 0 is the baseline market maker, w = 1 is pure model.



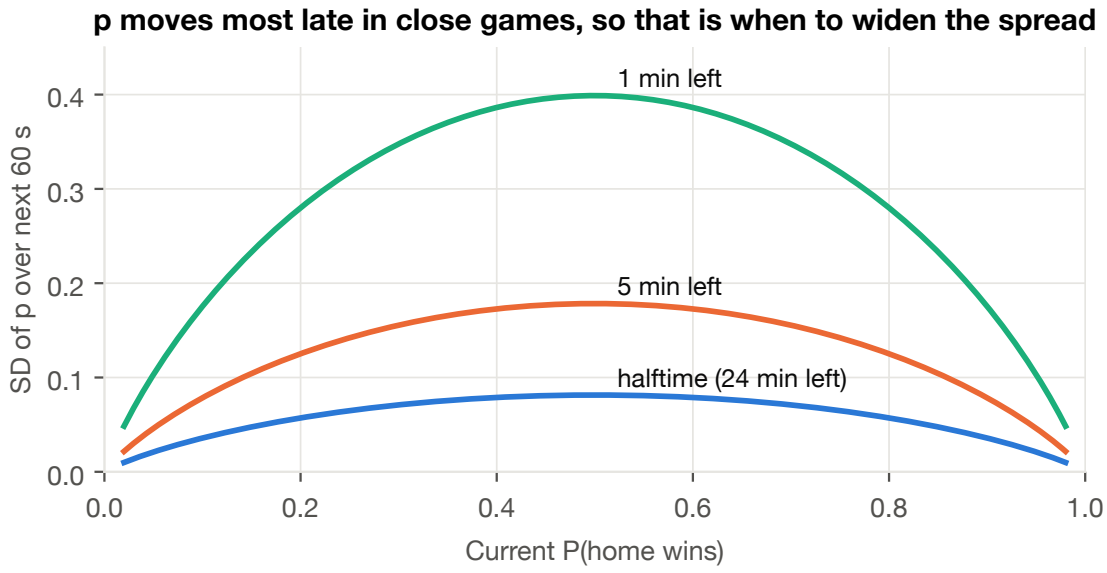
Top: no position, quotes symmetric around p. Bottom: after buying 40 contracts, the center leans down so the ask is more attractive and the bid less, nudging the position back toward zero.

Where the spread width comes from

Under the same Brownian model, how far p is likely to move in the next Δt seconds is

$$\mathrm{SD}(\Delta p) \approx \varphi(z) \cdot \sqrt{\Delta t / \text{seconds left}}, \quad z = \Phi^{-1}(p)$$

ϕ is the normal density, largest at p = 0.5. So the expected move is biggest when the game is close and nearly over, which is exactly when a market maker with stale quotes gets picked off. The spread should scale with it.



Expected one-minute move in p. At halftime a coin-flip game moves about 0.08 per minute; with one minute left it can move 0.4.

7. The backtest: how we know if it worked

We replay every Kalshi trade in the test period against our quotes. A trade "fills" us if a taker crossed our price. Two fill rules bracket the truth: **optimistic** (a trade at or through our price fills us) and **conservative** (only trades strictly through). We only claim results that survive the conservative one. We also feed the model the play-by-play 0, 5, 15 and 30 seconds late, because in real life the market often moves before the official feed does.

Report	What it tells us
PnL per game	Did the forecast-centered market maker make more than the mid-centered one? With confidence intervals from resampling games.
Spread earned	Income from being hit on both sides.
Adverse selection (60 s markout)	How much the price moved against us in the minute after each fill. The cost of trading with people who knew more.
Max drawdown	Worst peak-to-trough loss. The tail-risk number.
By window	Calm periods vs. the 2 minutes after a 6+ point run vs. the last 5 minutes of a close game. Fixed before we look at results.

The 2×2 design, center (mid vs. model) × spread (fixed vs. volatility-based), separates the two things we're claiming: a better center, and a smarter spread.

8. Your part, concretely

OWNED BY BRYAN

Live win probability model and calibration; quoting rule and tuning. In order:

1. **Build the training table** from `data/raw/pbp/` (downloaded, ~12.3k games). One row per 15 s: margin, seconds left, p_{pre}, home won. Use a stand-in prior until Arjav's Elo lands. (Due 11/5)
2. **Fit σ** by minimizing log loss; fit it per season too, and let 2024–25 decide how to weight old seasons.
3. **Logistic benchmark** (Yeh et al. features) and the Brier / log loss / calibration report vs. the Kalshi mid.
4. **Quoting rule**, one piece at a time, tuned on 2025 playoffs–Jan 2026. (Due 11/10)
5. **Out-of-sample run** with the team. (Due 11/12)

What you can safely ignore

- Scraping Kalshi and matching tickers to game IDs (Angel). You consume a `book` and `trades` table.
- Elo, closing lines, Pinnacle, injury PDFs (Arjav). You consume a `prior` table: game_id → p_{pre}.
- Polymarket. Dropped to optional in the Milestone I update.

Glossary

Calibrated	When the forecast says 70%, it happens 70% of the time.
Vig / de-vigged	A sportsbook's two sides sum to more than 100%; that excess is the vig. De-vigging rescales them to sum to 100%.
Elo	A team rating that rises when you beat good teams, falls when you lose to bad ones. Converts to a win probability.
Markout	Price change some fixed time after a fill. Negative markout means the taker was right and we were picked off.
Drawdown	Drop from a running high in cumulative PnL.
Out of sample	Data the model never saw while being fitted or tuned. The only results that count.
Favorite-longshot bias	The crowd pays too much for unlikely outcomes and too little for likely ones.
Inventory	Net contracts we hold. Positive = long = we profit if YES wins.

Sources: the team's one-pager and Milestone I update; Stern (1994); Avellaneda & Stoikov (2008); Yeh, Rice & Dubin (2022); Bürgi, Deng & Whelan (2025). Season figures computed from `shufinskiy/nba_data` play-by-play.